

Jongmin Moon

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RESEARCH KEYWORDS

Computational Neuroscience · Systems Neuroscience · Vision · Decision-making · Probabilistic Models

ACADEMIC APPOINTMENTS

2024 – present **Postdoctoral Fellow**
Center for Perceptual Systems, The University of Texas at Austin, TX, United States
• Advisor: Robbe L. T. Goris

EDUCATION

2018 – 2024 **Ph.D.**, Biomedical Engineering
Ulsan National Institute of Science and Technology, South Korea
• Advisor: Oh-Sang Kwon
• Thesis title: *Common computational principles underlying biases in visual perception, memory and confidence*

2022 Computational Neuroscience: Vision
Cold Spring Harbor Laboratory, NY, United States

2021 Deep Learning
Neuromatch Academy

2012 – 2018 **B.S.**, Human Factors Engineering & System Design and Control Engineering
Ulsan National Institute of Science and Technology, South Korea
• Compulsory military service in the Republic of Korea Air Force (2013 – 2015)

PUBLICATIONS

Peer-Reviewed Publications

- **Moon, J.**, Tadin, D.^{*}, & Kwon, O.-S.^{*} (2023). A key role of orientation in the coding of visual motion direction. *Psychonomic Bulletin & Review*, 30(2), 564–574. <https://doi.org/10.3758/s13423-022-02181-2>
- **Moon, J.**, & Kwon, O.-S. (2022). Attractive and repulsive effects of sensory history concurrently shape visual perception. *BMC Biology*, 20, 247. <https://doi.org/10.1186/s12915-022-01444-7>
- **Moon, J.**[#], Choe, S.[#], Lee, S., & Kwon, O.-S. (2019). Temporal dynamics of visual attention allocation. *Scientific Reports*, 9, 3664. <https://doi.org/10.1038/s41598-019-40281-7>

Manuscripts Under Review and in Preparation

- Saftari, L. N.[#], **Moon, J.**[#], & Kwon, O.-S. (in revision). Environmental motion presented ahead of self-motion modulates heading direction estimation.
- **Moon, J.**, Chung, D., Kim, H. J., Sul, S., & Kwon, O.-S. (in preparation). Optimal confidence judgments in perceptual estimations.
- **Moon, J.**, Yoon, H., & Kwon, O.-S. (in preparation). A behavioral signature of efficient coding in working memory representations.

CONFERENCE PRESENTATIONS

Conference Presentations

- Yoon, H., **Moon, J.**, & Kwon, O.-S. (2025, May 16–20). Attractive serial dependence occurs after motion repulsion. [Poster presentation]. *Vision Sciences Society 2025 Annual Meeting*, St. Pete Beach, FL, United States.
- **Moon, J.**, Yoon, H., & Kwon, O.-S. (2024, May 17–22). Probing bidirectional serial dependence in an N-back orientation estimation task. [Oral presentation]. *Vision Sciences Society 2024 Annual Meeting*, St. Pete Beach, FL, United States.
- **Moon, J.**, Saftari, L. N., & Kwon, O.-S. (2023, August 24–27). Environmental motion presented ahead of self-motion modulates the heading direction estimation. [Poster presentation]. *2023 Conference on Cognitive Computational Neuroscience*, Oxford, United Kingdom.
- Saftari, L. N., **Moon, J.**, & Kwon, O.-S. (2023, May 19–24). Synchronicity of visual and vestibular signals modulates the causal inference in heading direction estimation. [Poster presentation]. *Vision Sciences Society 2023 Annual Meeting*, St. Pete Beach, FL, United States.
- **Moon, J.**, & Kwon, O.-S. (2022, May 13–18). Confidence in perceptual estimation reflects behavioral variability, but not biases. [Poster presentation]. *Vision Sciences Society 2022 Annual Meeting*, St. Pete Beach, FL, United States.
- **Moon, J.**, & Kwon, O.-S. (2020, June 19–24). Motion perception is biased toward the orientation, not the direction, of the preceding motion. [Poster presentation]. *Vision Sciences Society 2020 Annual Meeting*, online.
- **Moon, J.**, & Kwon, O.-S. (2019, May 17–22). Additivity of attractive and repulsive sequential effects in motion direction estimation. [Oral presentation]. *Vision Sciences Society 2019 Annual Meeting*, St. Pete Beach, FL, United States.
- Choe, S., **Moon, J.**, & Kwon, O.-S. (2018, May 18–23). Temporal dynamics of visual attention allocation. [Poster presentation]. *Vision Sciences Society 2018 Annual Meeting*, St. Pete Beach, FL, United States.

Domestic Conference Presentations

- Yoon, H., **Moon, J.**, & Kwon, O.-S. (2025, February 12–14). Attractive serial dependence occurs after motion repulsion. [Poster presentation]. *62nd Annual Meeting of the Korean Society of Cognitive and Biological Psychology*, Gwangju, South Korea.
- **Moon, J.**, Yoon, H., & Kwon, O.-S. (2024, February 1–3). Probing bidirectional serial dependence in an N-back orientation estimation task. [Oral presentation]. *61st Annual Meeting of the Korean Society of Cognitive and Biological Psychology*, Seoul, South Korea.
- **Moon, J.**, Saftari, L. N., & Kwon, O.-S. (2023, February 16–17). An environmental motion presented ahead of body motion modulates the causal inference in heading direction estimation. [Oral presentation]. *60th Annual Meeting of the Korean Society of Cognitive and Biological Psychology*, Seoul, South Korea.
- **Moon, J.**, & Kwon, O.-S. (2021, July 15–16). The effects of behavioral variability and biases on confidence in perceptual estimation. [Oral presentation]. *58th Annual Meeting of the Korean Society of Cognitive and Biological Psychology*, online.
- **Moon, J.**, & Kwon, O.-S. (2020, August 27–28). A key role of non-directional orientation in the coding of visual motion direction. [Oral presentation]. *57th Annual Meeting of the Korean Society of Cognitive and Biological Psychology*, online.
- **Moon, J.**, & Kwon, O.-S. (2019, February 14–15). Sequential effects in motion direction estimation. [Oral presentation]. *56th Annual Meeting of the Korean Society of Cognitive and Biological Psychology*, Pyeongchang, South Korea.
- Choe, S., **Moon, J.**, & Kwon, O.-S. (2018, January 25–26). Temporal dynamics of visual attention. [Poster presentation]. *55th Annual Meeting of the Korean Society of Cognitive and Biological Psychology*, Suwon, South Korea.

HONORS AND AWARDS

2024	Best Presentation Award	Korean Society of Cognitive and Biological Psychology
2024	2nd Prize in Data Challenge	Korean Society of Cognitive and Biological Psychology
2023	Best Presentation Award	Korean Society of Cognitive and Biological Psychology
2022	CSHL Course Award	Howard Hughes Medical Institute
2021	Best Presentation Award	Korean Society of Cognitive and Biological Psychology
2018 – 2024	UNIST Graduate Scholarship	Ulsan National Institute of Science and Technology
2018	Best Poster Award	Korean Society of Cognitive and Biological Psychology
2016	Samsung Dream Class Scholarship	Samsung Welfare Foundation
2012	UNIST Dream Scholarship	Ulsan National Institute of Science and Technology
2012 – 2018	UNIST Undergraduate Scholarship	Ulsan National Institute of Science and Technology

TEACHING

Teaching Assistant

2019 – 2023	Experimental Design	Department of Biomedical Engineering, UNIST
2019 – 2022	Sensation and Perception	Department of Biomedical Engineering, UNIST
2017 – 2018	Science of Human Behavior	School of Liberal Arts, UNIST

Other Experiences

2016	IT Instructor	World Friends ICT Volunteer Program (in Ghana)
2016	Mentor	Samsung Dream Class, Samsung Welfare Foundation
2014	Mentor	Educational Outreach Program, ROKAF
2013 – 2015	Assistant Drill Instructor	Republic of Korea Air Force

ACADEMIC SERVICE

Ad Hoc Journal Reviewer

Nature Communications
Psychological Research

Ad Hoc Conference Reviewer

Conference on Cognitive Computational Neuroscience

REFERENCES

Robbe L. T. Goris, Ph.D. Associate Professor Center for Perceptual Systems The University of Texas at Austin robbe.goris@utexas.edu	Oh-Sang Kwon, Ph.D. Associate Professor Department of Biomedical Engineering Ulsan National Institute of Science and Technology oskwon@unist.ac.kr
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